

Fraunhofer Italia – Leonardo Venturoso

Real-time Traffic Prediction in Bolzano Using Bluetooth Sensor Data: A Big Data Approach

***Innovations for Transport and Sustainable Mobility
Workshop, 03-04/04/2025, NOI Techpark, Bruneck***



unibz

Greetings

A few words about myself

Education:

MSc in Data Science

University of Trento, Italy (2020-2023)

MSc in Cognitive Neuroscience

University of Trento, Italy (2016-2018)

BSc in Cognitive Science

University of Trento, Italy (2013-2016)

Academic Experience:

Post Graduate Researcher and Python Developer

CiMeC, Italy (2019-2020)

Graduate Visiting Researcher

University of Toronto, Canada (Aug 2018-Dec 2018)

Visiting Researcher

Nanyang Technological University, Singapore (May 2016-Aug 2016)



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Work Experience:

Data Scientist

Fraunhofer Italia IEC, Bolzano, Italy (Feb 2024-Ongoing)

Data Science Researcher (MSc Thesis)

Fondazione Bruno Kessler, Trento, Italy (Mar 2023- Mar 2024)

Data Scientist

AlmaViva SpA, Rovereto, Italy (Jan 2023-Jan 2024)

Data Analyst and Software Developer

Edizioni Centro Studi Erickson, Trento, Italy (Jan 2021-Jan 2023)



My activities and projects in Fraunhofer Italia as a Data Scientist

Industrial Projects on:

A) Artificial Intelligence for Quality Control



B) Optimization for Flexible Production Systems and Logistics

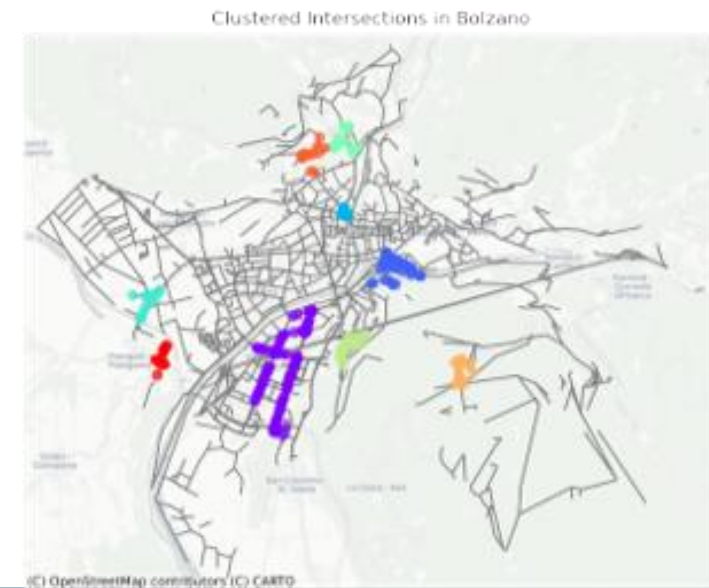
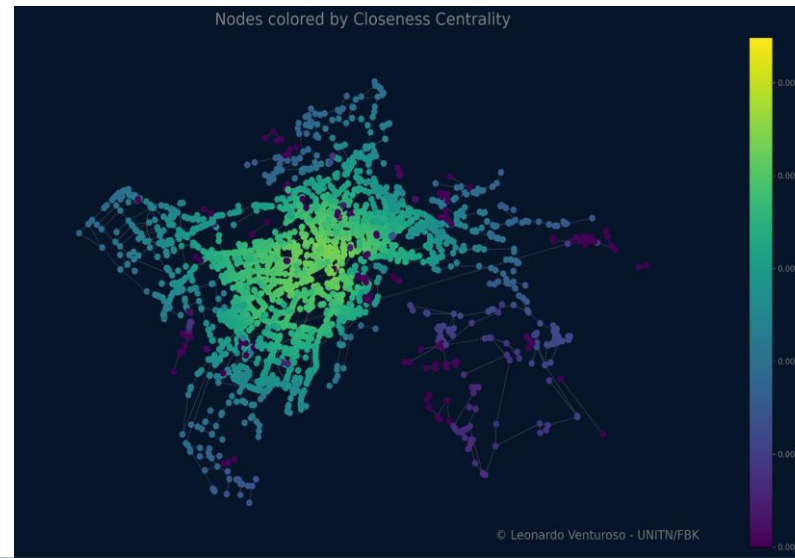
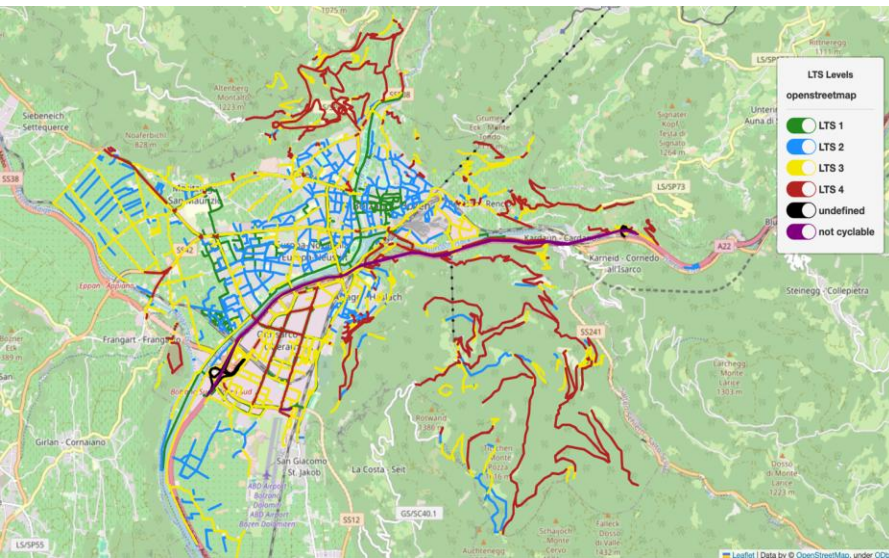


Bioeconomy and Sustainability activities

NEW TOPIC:

C) Artificial Intelligence and Data Driven Solutions for Sustainable Mobility

Exploring the intersection of artificial intelligence and sustainable transportation to develop greener, more efficient mobility solutions => Municipality of Bolzano Prize 2024 on Level of Traffic Stress and Infrastructure Network Analysis for Safe and Accessible Cycling Project. Paper under review.



AI for Mobility: Real-time Traffic Prediction

Tumminello A, Venturoso L, (2021) – University of Trento



Background:

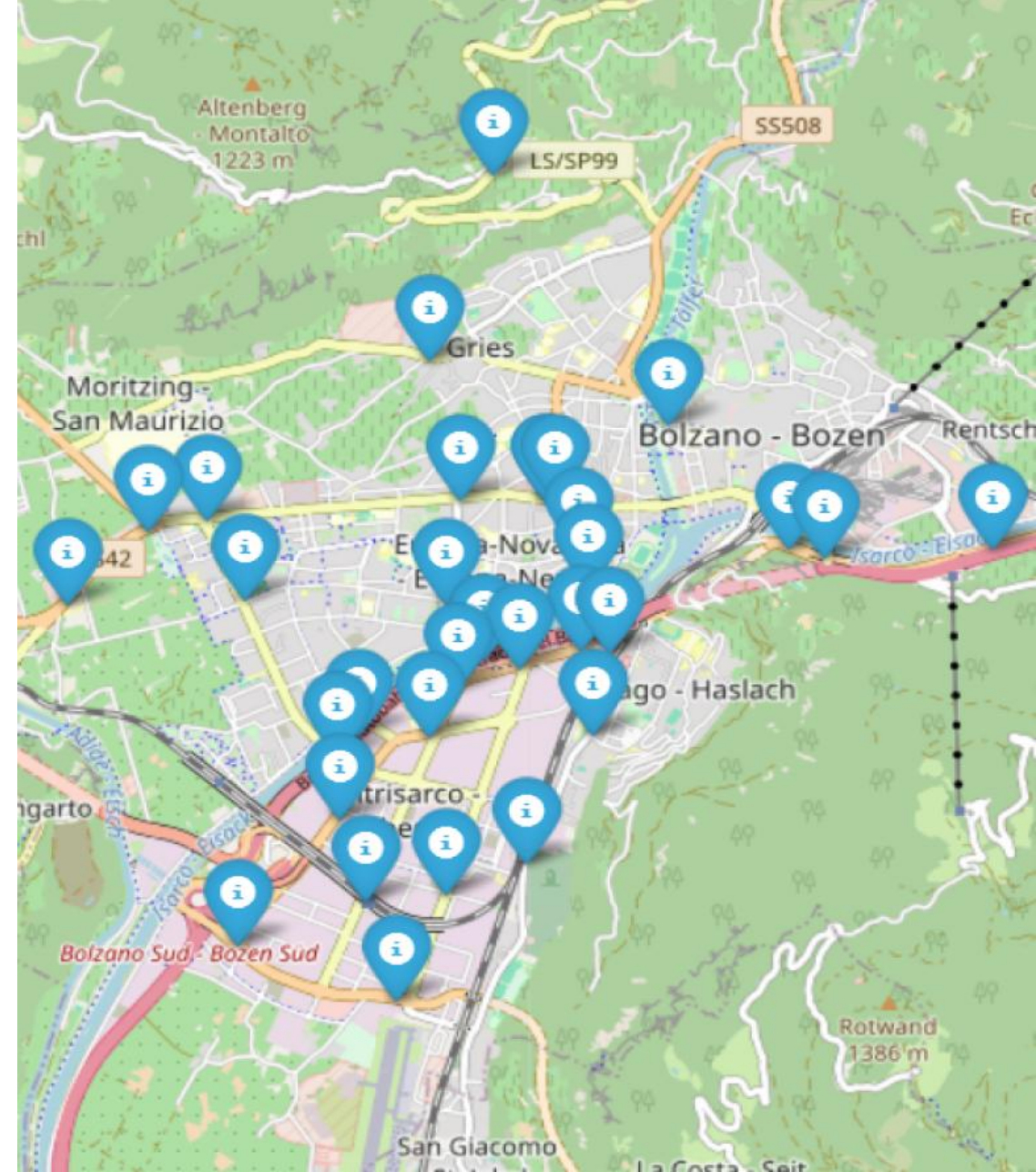
- **Traffic congestion** impacts a country's economy and health.
- Urban areas struggle with efficient traffic flow, making **real-time data crucial for smart mobility and livability** (Guo et al., 2021).
- **Data** sources include **stationary (sensors, cameras) and probe (GPS)** data.
- Various models (e.g., ARIMA, SVR, RFR, NN, DNN, LSTM) predict traffic and account for external factors like weather, construction, and accidents (Akhtar & Moridpour, 2021).

Research goals and questions

Aim

The purpose of our project was to design and implement a **big data system** able to **predict in near-time the traffic** associated with specific and crucial points of interest monitored by **Bluetooth sensors in the Bolzano area**, using **Open Data Hub** databases.

OPEN DATA HUB



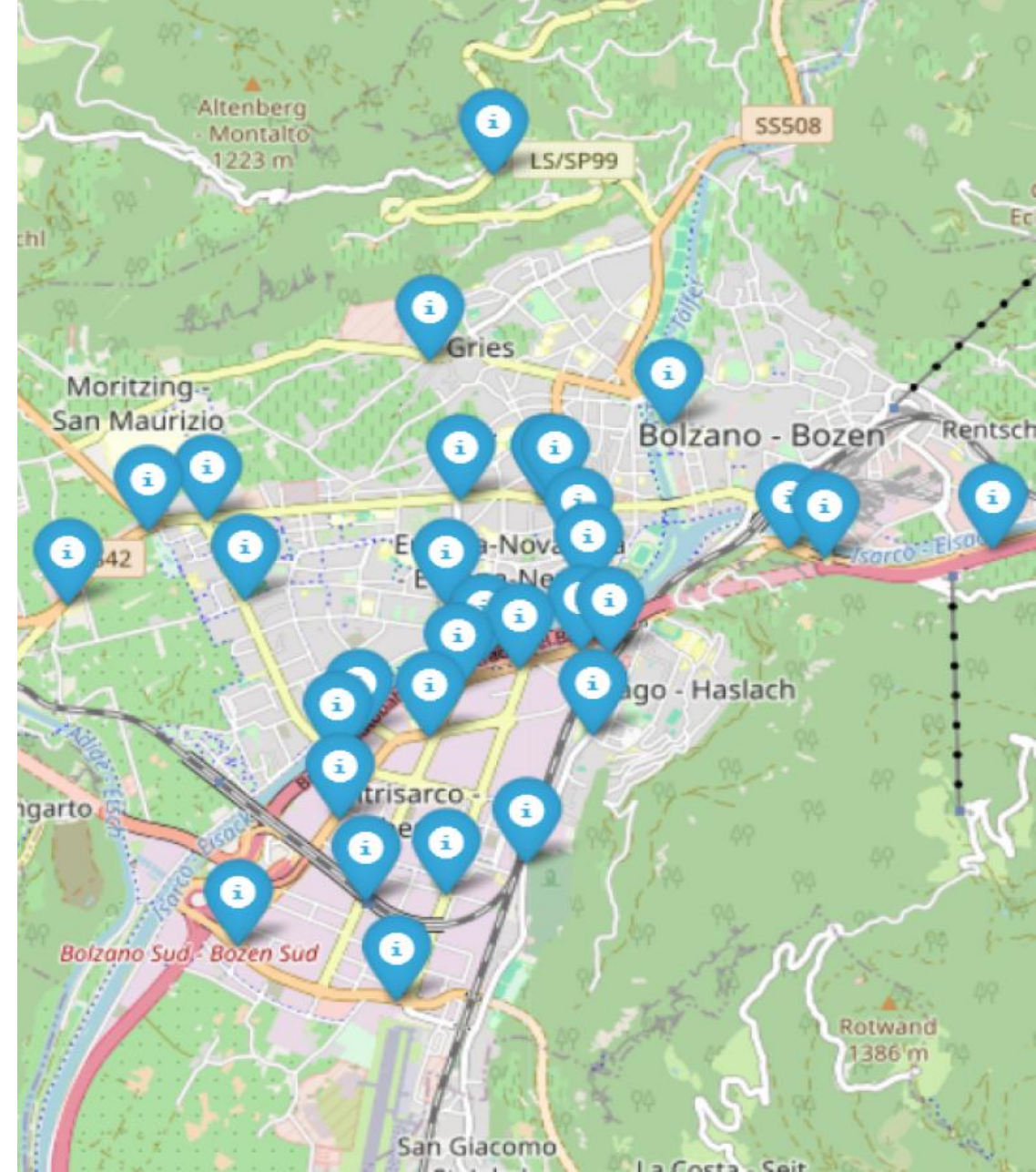
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Interesting but...how?

OPEN DATA HUB



System Model

Overview of the problem:

Conceptual:

the problem is to predict the traffic (i.e. The number of vehicles detected through Bluetooth sensors stations in near time. The number of stations increase gradually over time.

System Model

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Conceptual:

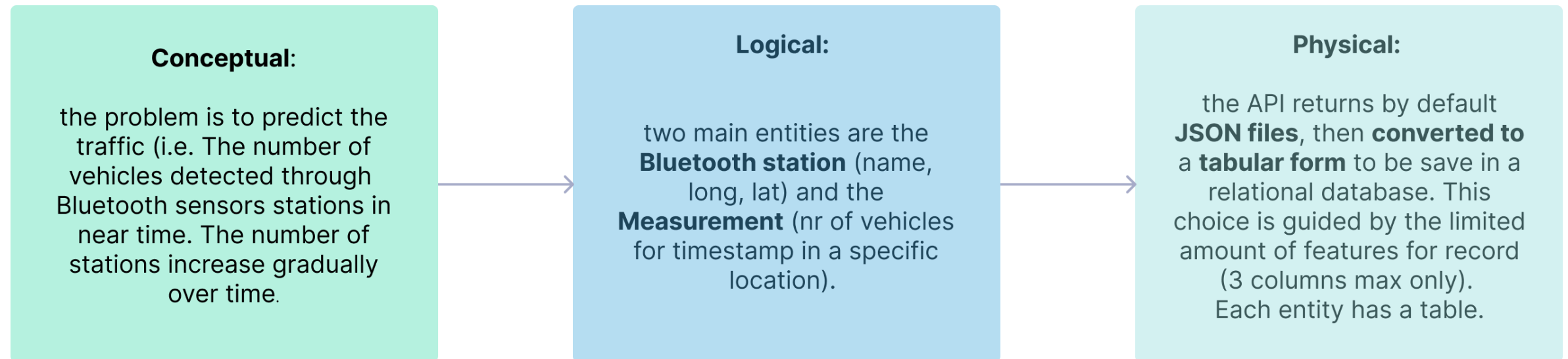
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Logical:

two main entities are the **Bluetooth station** (name, long, lat) and the **Measurement** (nr of vehicles for timestamp in a specific location).

System Model

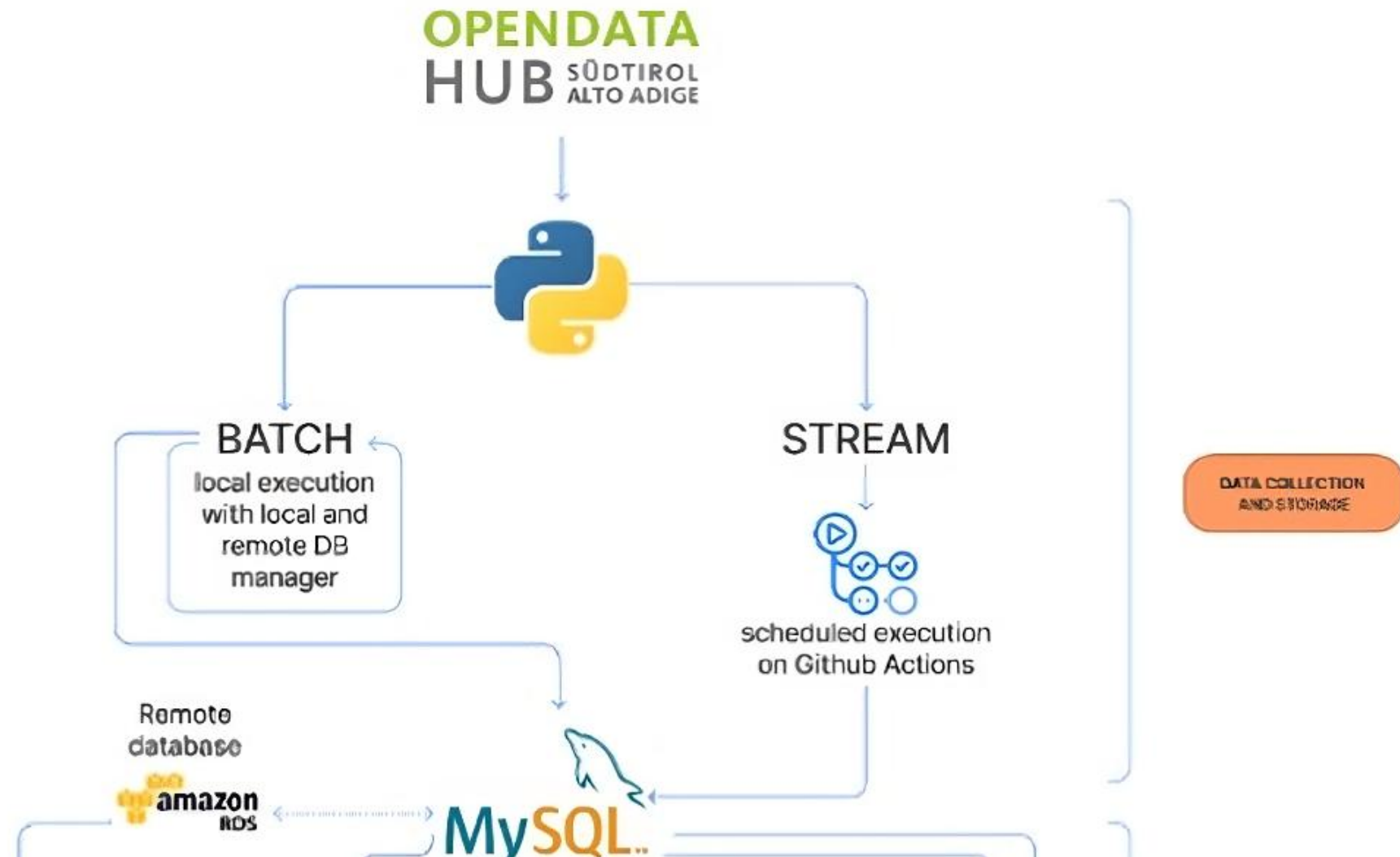
Overview of the problem:



Methods

Main Steps:

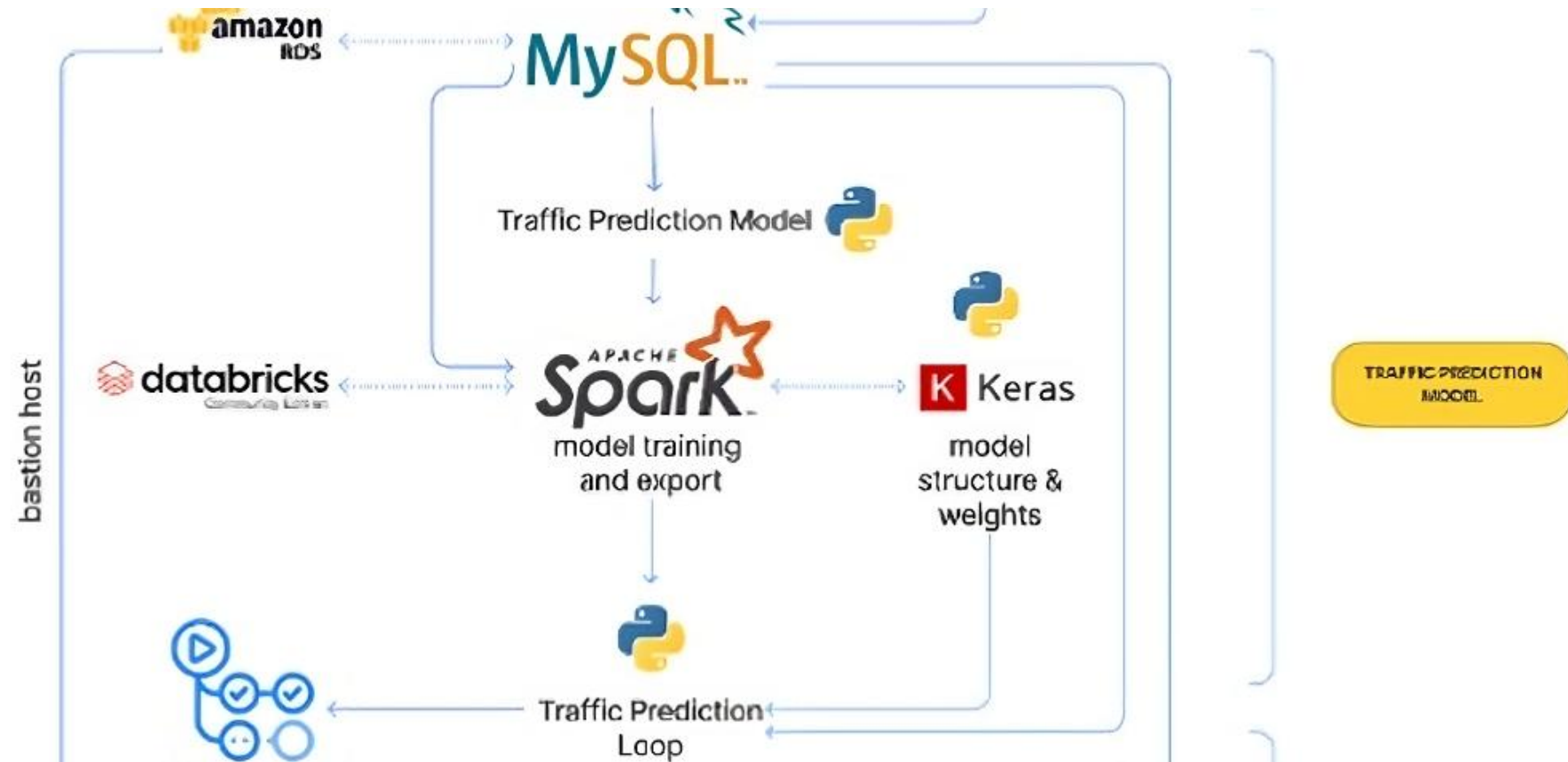
- 1) Data Collection and Automation
- 2) Data Ingestion
- 3) Model Implementation
- 4) Data Visualization and Application Development



Methods

Main Steps:

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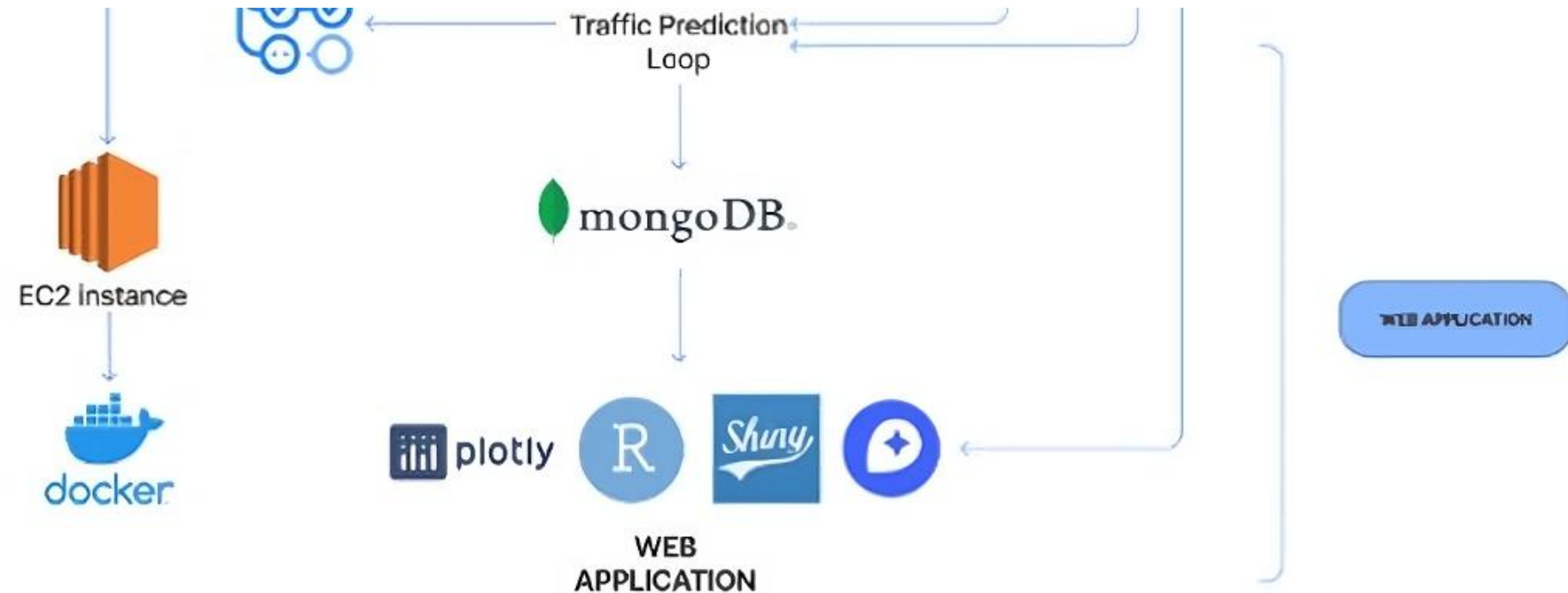


Figure 1: Traffic Data Pipeline

Results:

Prediction Results:

- Separated document for different stations and timestamps where **predictions** are **saved in MongoDB**.
- **Model updated** with these predictions, then repeat the process by merging the new predictions into the original dataset. This cycle, running **every two hours**, allows for **forecasts up to 10 hours ahead**, helping traffic management and planners maintain timely and actionable forecasts.

Results:

Web app with:

- **User input** (theme, date, station) **and output** (plots, tables).
- **Bubbles** in correspondence of geographical coordinates of Bluetooth Stations. **Statistics** (date, hour, weekday, month) and **grouping** based on station and hour.



Results: ShinyApp

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Search Bluetooth Station

MENU

Real time traffic

Stations

Historical Traffic

Compare

OTHERS

Settings

Read Me

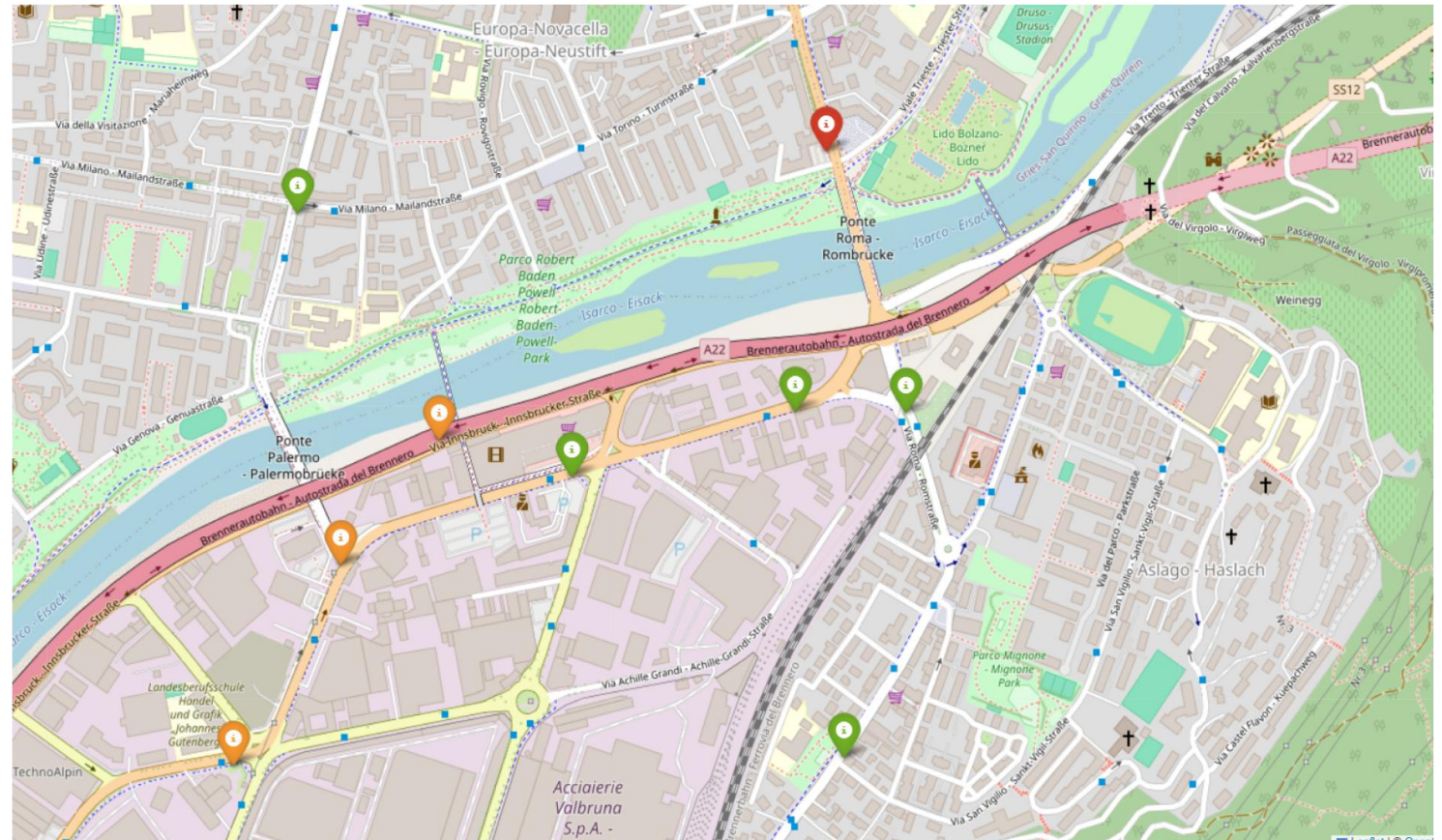
Traffic in Bolzano

Real time traffic

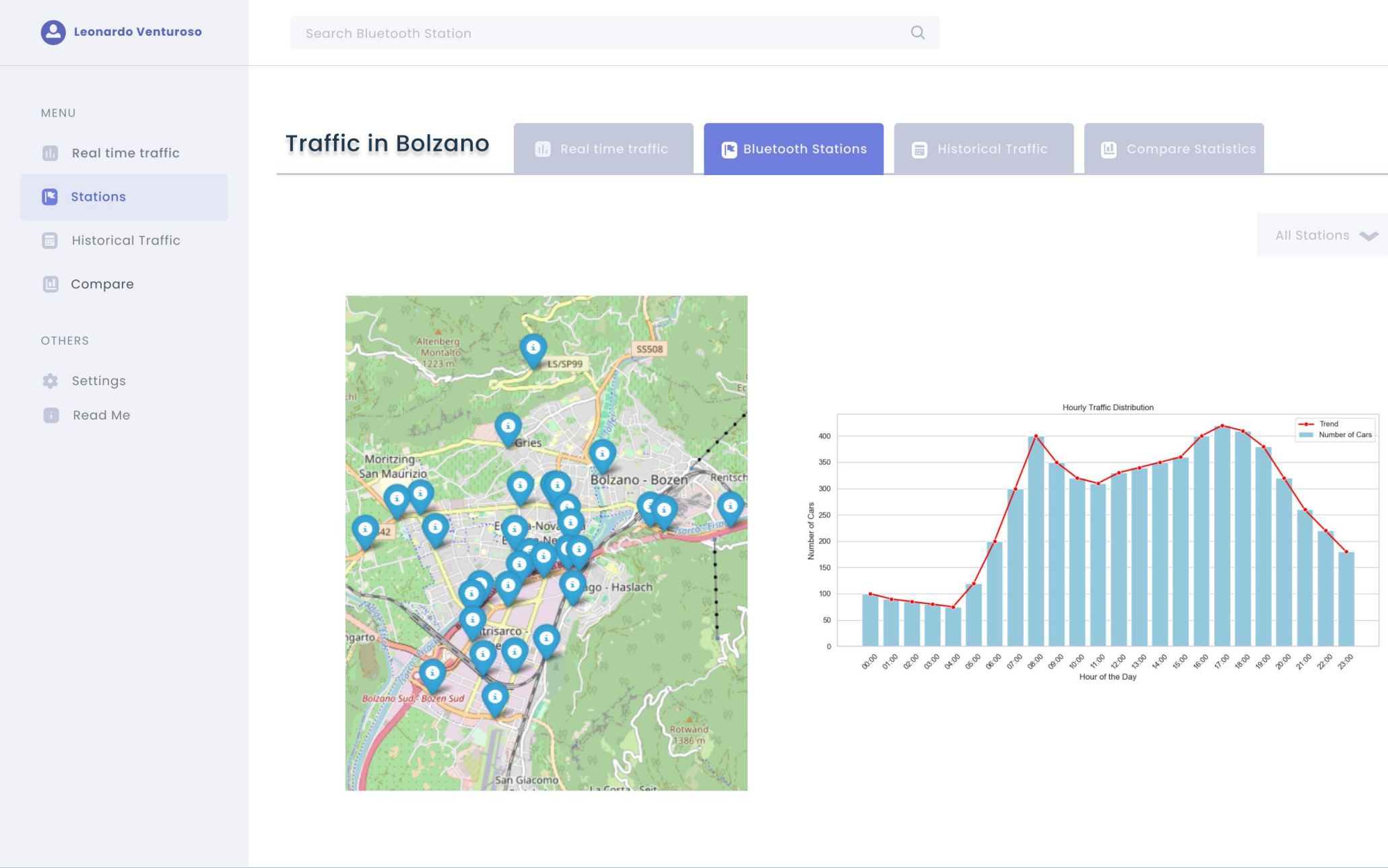
Bluetooth Stations

Historical Traffic

Compare Statistics



Results: ShinyApp



Results: ShinyApp

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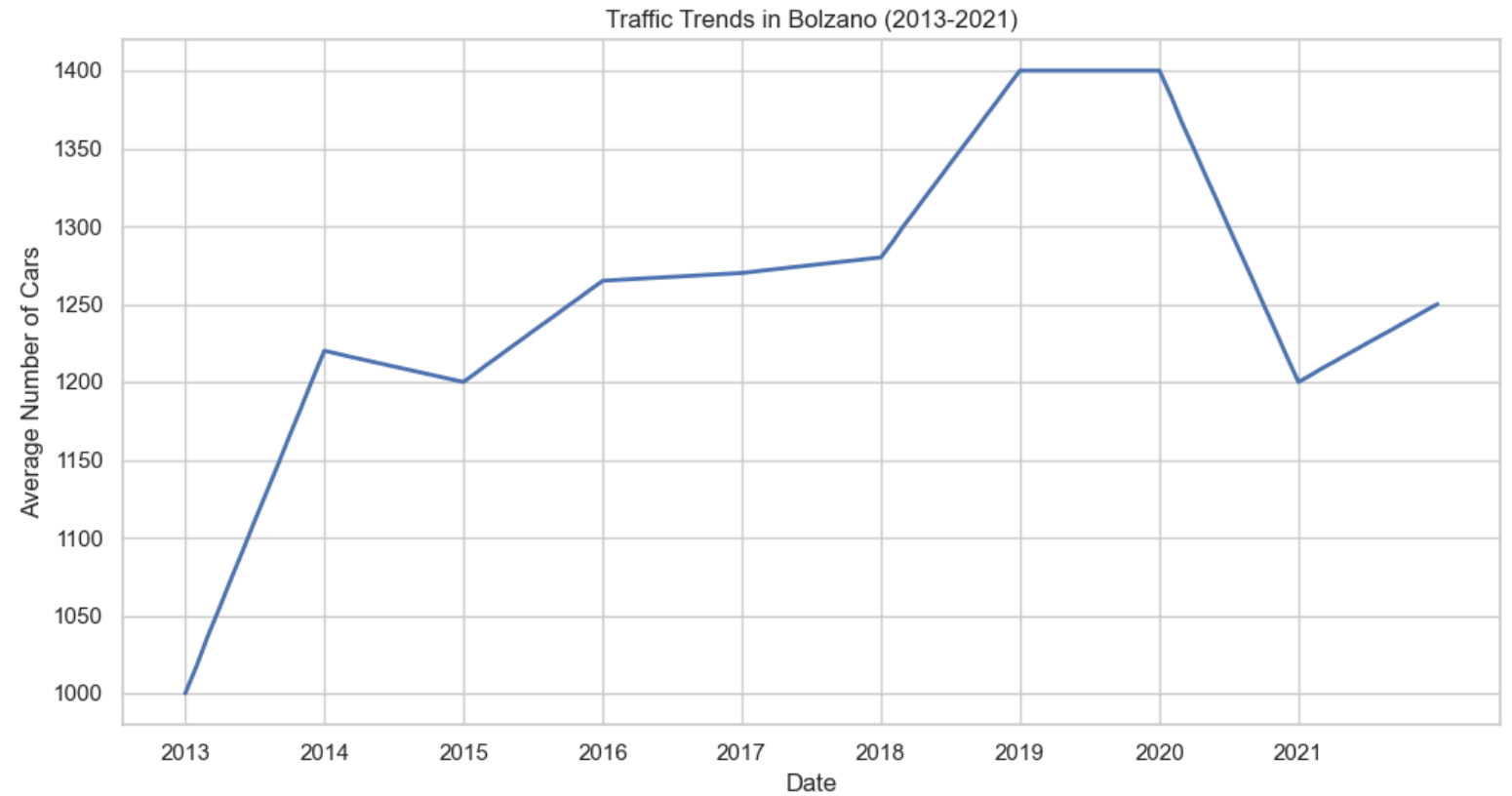
Real time traffic

Bluetooth Stations

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2013 - 2021



Results:

ShinyApp

Leonardo Venturoso

Search Bluetooth Station



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Traffic in Bolzano

Real time traffic

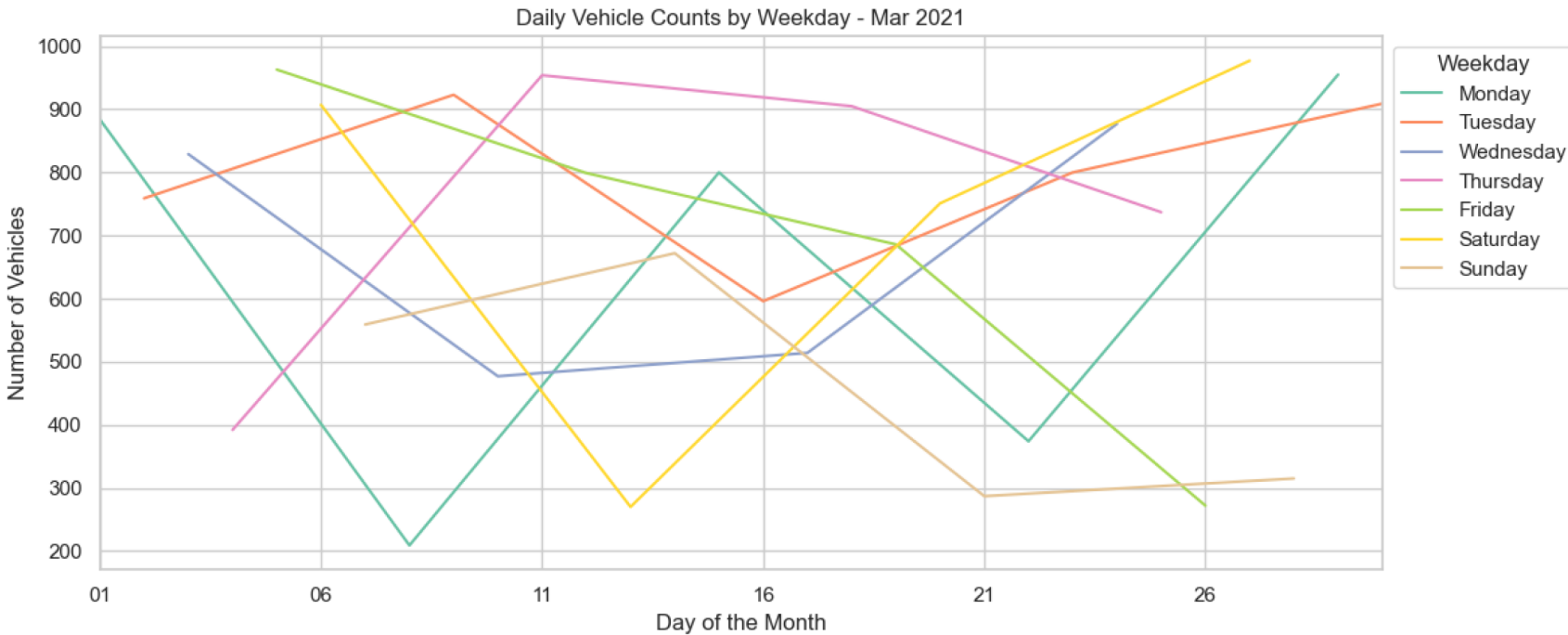
Bluetooth Stations

Historical Traffic

Compare Statistics

All weekdays

Select Month



Limitations and Future Directions

Some limitations:

- **More data** can be helpful (more stations).
- Script **automation** could have been **scheduled more frequently**.
- **MySQL may not be the best choice**.
- LSTM relies on historical data, data processing tailored specifically for this model, **resource-intensive process**.
- **Other factors to consider**: location, time, seasonality to enhance accuracy.
- **App framework** (Shiny is slow but free)

Let's get in touch for future collaborations!

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Thank you for your attention!
